Stash: setting work aside

Objectives

- Explain what git stash saves and what it leaves behind by default
- Use git stash, git stash list and git stash pop to switch tasks and come back

Teaching

Start with a guess

Sometimes you must drop what you are doing: an urgent bug needs you on another branch. Committing half-done work just to switch is messy. Git can set it aside instead.

The stash is a shelf for unfinished work

git stash saves your changes to tracked files (staged or not) on a stack called the stash, and resets those files to the last commit. Now you can switch branches.

By default untracked files, never added, stay in the working copy; git stash -u includes them.

git stash list shows entries; the newest is stash@{0}. git stash pop reapplies the newest entry to the branch you are on now and, once it applies cleanly, removes it from the stack, so switch back first. If applying it causes conflicts, the entry stays in the stash: resolve them, then remove it with git stash drop.

Apply it to new files

Same rule: plain git stash takes tracked changes only.

Model

Practice repository Simulation: no real Git runs

Working copyon feature

- app.jsmodified
- draft.mduntracked

Staging areanext commit

- nothing staged

Brancheslatest commit

- mainC3 Fix login
- HEAD → featureC4 Add search

Stashnewest first

- empty

Recap

- git stash sets aside changes to tracked files and returns them to the last commit, so you can switch.
- Untracked files stay unless you use git stash -u.
- git stash list shows entries; stash@{0} is the newest.
- git stash pop reapplies the newest entry to the current branch and removes it once it applies cleanly (after a conflict the entry stays until you drop it), so switch back first.

Next topic: Remotes: SSH keys, clone, push, fetch vs pull

Cheat sheet

git stash	set aside tracked changes
git stash -u	also set aside untracked files
git stash list	show saved entries
git stash pop	reapply newest entry here and remove it (kept if it conflicts)

Sources

Teaching claims were checked on 2026-10-06 (UTC) against live official documentation: the Git reference manual on git-scm.com (latest source release 2.56.0, released 2026-09-28). Simulated command output is illustrative: its exact wording was not checked and varies with Git version and settings.

- git-stash: description

- git-stash: options (--include-untracked)
- git-switch: description

Static interaction alternatives

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predict

Prediction (not scored). You have an unfinished edit to app.js on feature. You run git stash. What happens to the edit? Options: Saved on a stack; the file goes back to the last commit; Committed to feature; Deleted; Moved to main. Answer: Saved on a stack; the file goes back to the last commit. Saved on a stack; the file goes back to the last commit: git stash saves changes to tracked files on a stack called the stash and cleans them out of your working copy. Committed to feature: Attractive, but stash makes no commit on your branch; the branch history does not change. Deleted: Nothing is lost: the change is kept in the stash until you bring it back. Moved to main: Stash is not tied to switching; it just sets the change aside.

terminal

Practice terminal (simulated; nothing runs on your computer). Task: Your app.js edit on feature is unfinished, and an urgent fix waits on main. Visit main, then return and resume. Goals: Set aside the app.js edit; Visit main; Back on feature, edit restored, stash empty. One sequence that meets every goal: git stash, then git switch main, then git switch feature, then git stash pop.

mcq

Check (one retry, then the answer is shown). You have a modified tracked file cart.js and a new untracked file todo.txt. You run git stash. What does git status show next? Options: Only todo.txt, as untracked; Nothing: working tree clean; cart.js modified; Both cart.js and todo.txt. Answer: Only todo.txt, as untracked. Only todo.txt, as untracked: Right: the cart.js change went into the stash; untracked todo.txt stays because plain git stash skips untracked files. Nothing: working tree clean: That needs git stash -u, which also takes untracked files. cart.js modified: The tracked change to cart.js is exactly what stash takes away. Both cart.js and todo.txt: Stash did remove the cart.js change; only todo.txt remains. Retry question: The model shows style.css modified (tracked) and notes.md untracked. After plain git stash, where is each file's change? Options: Stash: style.css. Working copy: notes.md; Stash: style.css and notes.md; Stash: notes.md. Working copy: style.css; Neither; both are committed. Answer: Stash: style.css. Working copy: notes.md. Stash: style.css. Working copy: notes.md: Right: tracked change stashed, untracked file left behind. Stash: style.css and notes.md: Only git stash -u would take the untracked notes.md. Stash: notes.md. Working copy: style.css: Backwards: stash takes tracked changes, not untracked ones. Neither; both are committed: Stash never commits to your branch.